Lesson 7.4 Guided Notes

Session 1 · Comparing Numbers on a Number Line

I can compare two rational numbers using their positions on a number line. · 6.NS.C.7a

Word / Spanish	Meaning — complete the blank	Example
greater than (>) / mayor que	Farther to the _____ on the number line.	3 > −6
less than (<) / menor que	Farther to the _____ on the number line.	−6 < −2
inequality / desigualdad	A math sentence that _____ numbers using < or >.	−8 < −5

Key idea

On a number line, the number farther _____ is greater.

For two negative numbers, the one closer to _____ is greater.

Worked example

Compare the temperatures −14 degrees and −5 degrees.

−5 is to the _____ of −14.

So −14 _____ −5.

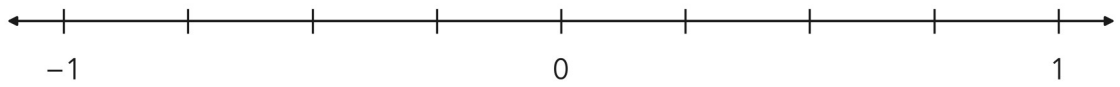
Answer: −5 is _____; −14 is colder.

You try

1. Write < or > in each blank.
a. −7 ___ −3 _____
b. −1.5 ___ −2.5 _____

_____

2. Plot −3/4 and −1/4. Write a true comparison using <.



_____

Sum it up: To compare two negative numbers, I ___ because ___.

_____

Session 2 · Ordering Integers and Rational Numbers

I can order integers and rational numbers from least to greatest. · 6.NS.C.7b

Word / Spanish	Meaning — complete the blank	Example
order / orden	To list numbers from _____ to greatest (or greatest to least).	−9, −2, 0, 6
least / mínimo	The smallest number; it is farthest to the _____.	−9 is the least
greatest / máximo	The biggest number; it is farthest to the _____.	6 is the greatest

Key idea

Least to greatest means _____ to right on a number line.

Write every number in the same _____ (such as decimals) before you order.

Worked example

Order −6, 1.5, −0.5, and 3/4 from least to greatest.

Write 3/4 as a decimal: _____.

Negatives first: −6 < _____.

Then positives: 0.75 < 1.5.

Answer: −6, −0.5, 3/4, _____

You try

1. Order from least to greatest.
a. −3, −8, 1 _____
b. −0.2, −2, −0.02 _____

_____

2. Three divers are at −12 m, −4.5 m, and −20.25 m. Which diver is deepest?

_____

Sum it up: To order rational numbers, I ___, then I ___.

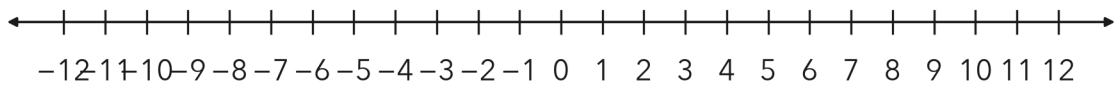
_____

Session 1 · Model, reason, and explain

Comparing Numbers on a Number Line · 6.NS.C.7a

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Compare temperatures −12°C and −4°C. Which is warmer? Which is farther from 0°C? Use a number line to justify both.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: assumes greater distance from zero means a greater negative number.. Explain what changes and what stays true.

_____

_____

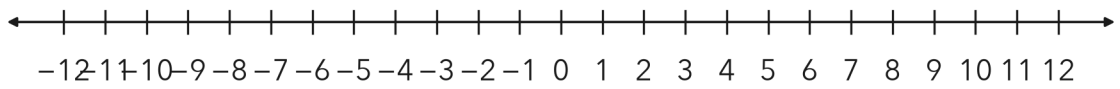
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Ordering Integers and Rational Numbers · 6.NS.C.7b

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Order −1.2, −3/4, 0, and −0.5 from least to greatest. Explain the ordering of the negative fraction and decimal.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: uses positive-number digit comparisons for negative numbers.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?